Airfoil Exploration using NASA Simulator

Project Report Submitted in fulfillment of the Requirements for the Award of the Internship of Winter Training Program on Space Science Technology

(Subject Name: Advanced Drone Technology)

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Airfoils:

Anairfoilis (also called as aerofoil) a streamlined body over the wing that is capable of generating significantly more lift than drag. The shape determines how lift and drag are generated.Lift is produced due to pressure differences created by the shape of the airfoil

Key Terms that determine the effect of airfoil:

1) Chord line: The chord line is the straight imaginary line connecting the leading edge (front) and trailing edge (back) of an airfoil. It's used as the reference line to measure angles like the angle of attack.

2) Angle of Attack: The angle between the relative wind and the chord line is the angle of attack on an airfoil.

Types of Airfoils:

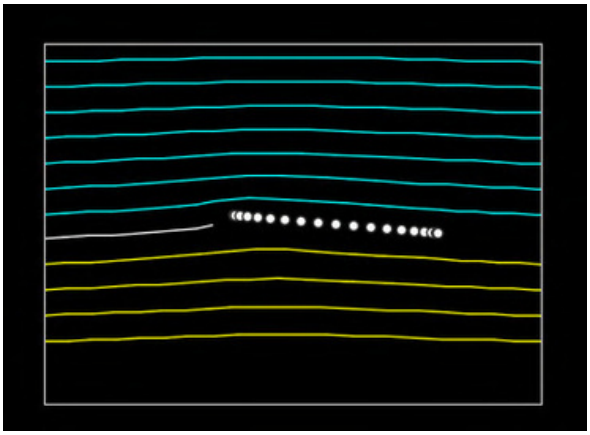
There arethreemajortypes of airfoils in aerodynamic studies:

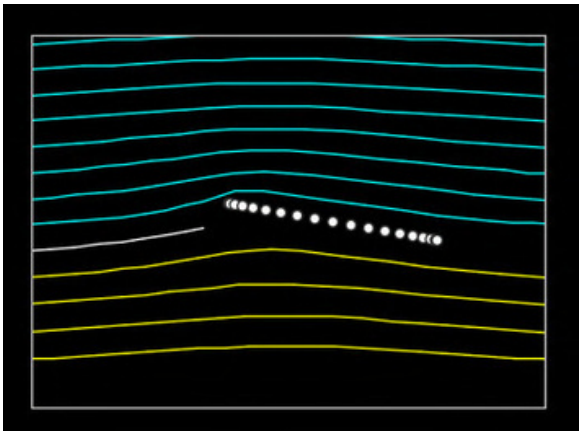
1) Flat type:

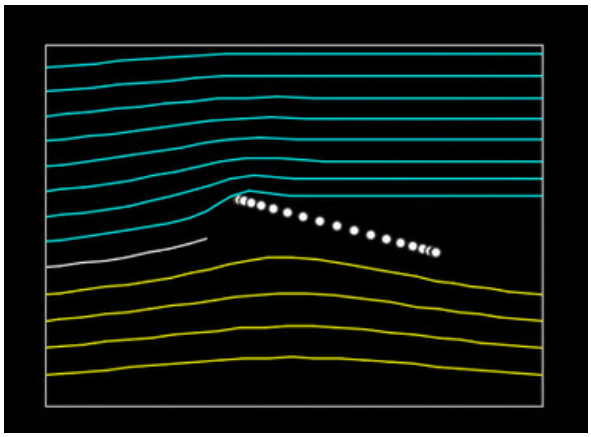
- A flat plate is the simplest shape used in basic aerodynamic studies.
- Even a flat plate can generate lift when it's set at a correct angle of attack .

Fig: 1

- (Fig 1) displays a flat airfoil held at zero degrees, resulting in higher drag due to the lack of aerodynamic lift because the airflow does not create a significant pressure difference between the upper and lower surfaces.







- The images above illustrate a flat-type airfoil simulated at angles of attack of 5 degrees, 10 degrees, and 15 degrees, respectively.

Angle of Attack	Lift (lbf)	Drag (lbf)
0 degree	0	23
5 degree	1342	122
10 degree	2568	384
15 degree	3269	672

Table: 1 Flat Type Analysis

2) Symmetric type:

- Symmetric airfoils have identical upper and lower surfaces, so the camber is zero.
- Lift increases only when the airfoil is tilted relative to the airflow

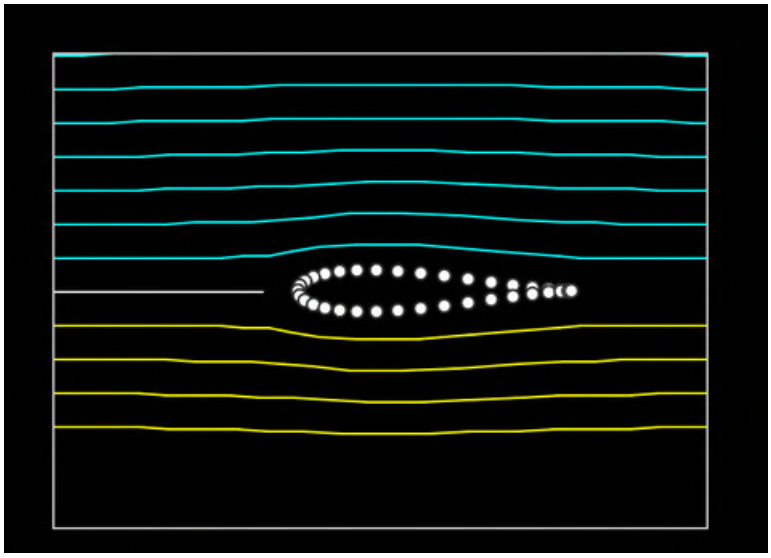
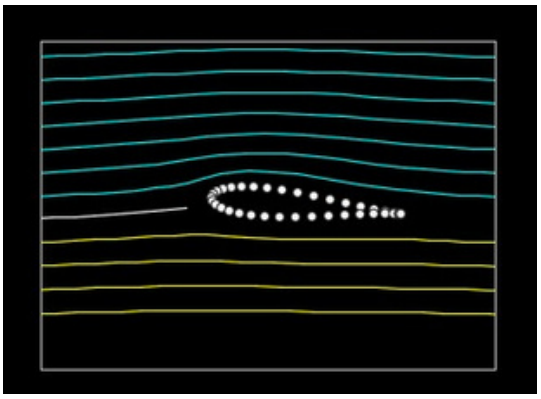
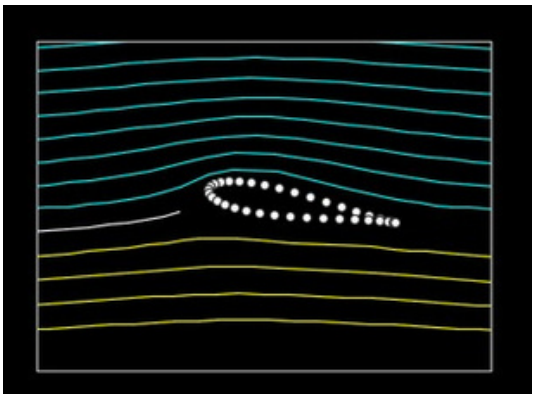
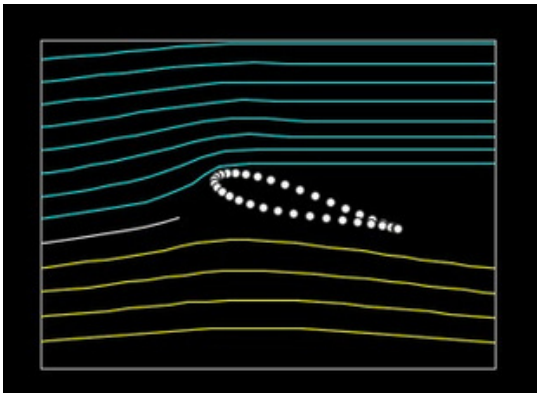


Fig: 2

- (Fig 2) displays a symmetric airfoil held at zero degrees angle of attack. Since the upper and lower surfaces are identical, air flows evenly on both sides of the airfoil, resulting in no lift being produced. However, drag is still present due to air resistance.







- The images above illustrate a Symmetric-type airfoil simulated at angles of attack of 5 degrees, 10 degrees, and 15 degrees, respectively.

Angle of Attack	Lift (lbf)	Drag (lbf)
0 degree	0	43
5 degree	1492	132
10 degree	2842	373
15 degree	3609	678

Table: 2 Symmetric Type Analysis

3) Cambered type:

- Cambered airfoils have curved upper and lower surfaces with different shapes.
- They can produce lift even at zero angle of attack because the curvature itself affects airflow.

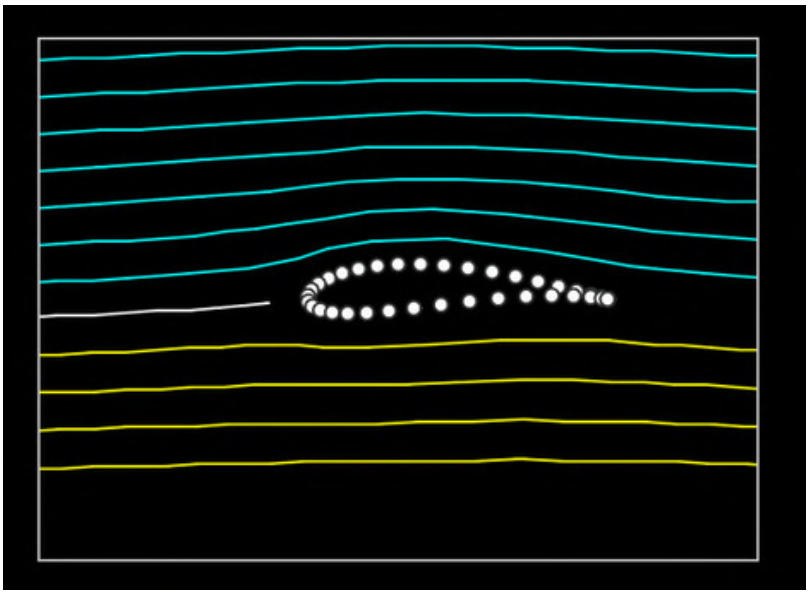
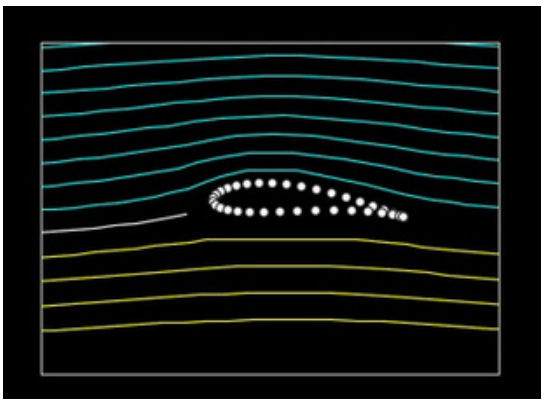
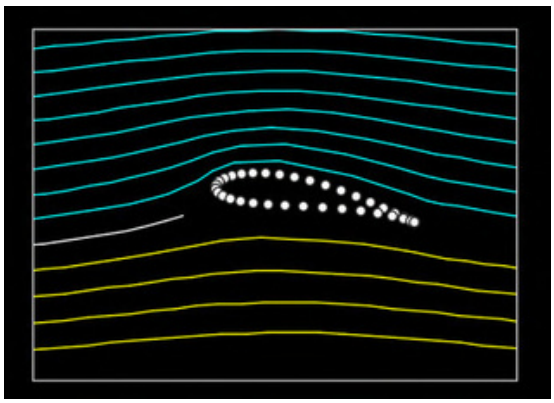
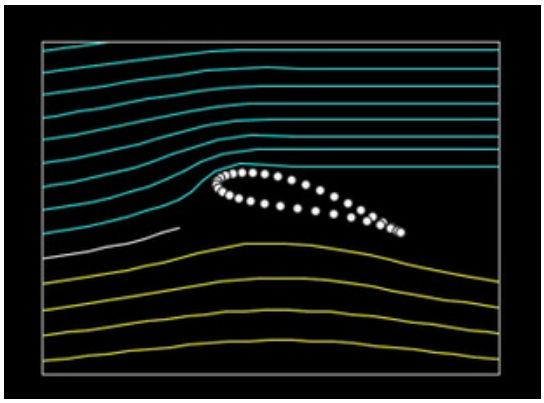


Fig: 3

- (Fig 3) displays a cambered airfoil with 5.5% camber held at zero degrees angle of attack. Due to the curved shape of the airfoil, air moves faster over the upper surface than the lower surface, creating a pressure difference. As a result, the airfoil produces lift even without being tilted, while drag remains relatively low.







- The images above illustrate a Cambered-type airfoil simulated at angles of attack of 5 degrees, 10 degrees, and 15 degrees, respectively.

Angle of Attack	Lift (lbf)	Drag (lbf)
0 degree	1653	145
5 degree	2994	416
10 degree	4203	803
15 degree	4726	1063

Table: 3 Cambered Type Analysis

Analysis:

At moderate angles of attack between 5° and 10°, the cambered airfoil produced the highest lift compared to symmetric and flat plate airfoils. This is because the curved geometry of the cambered airfoil accelerates airflow over the upper surface more effectively, creating a larger pressure difference and thus higher lift at lower angles.

When the angle of attack was increased to around 15°, the cambered airfoil stalled first, with a stall angle of approximately 14.4°. In contrast, both the symmetric and flat plate airfoils stalled later at around 15.5°. The earlier stall of the cambered airfoil occurs due to its higher curvature, which causes airflow separation to begin sooner at high angles of attack.

A symmetric airfoil shows nearly zero lift at 0° angle of attack because its upper and lower surfaces are identical. Under this condition, airflow splits evenly and travels at similar velocities on both sides, resulting in no significant pressure difference. Since lift is generated by pressure imbalance, the net lift becomes zero.

Cambered airfoils are capable of generating lift even at zero angle of attack because of their asymmetric shape. The camber causes airflow to move faster over the upper surface than the lower surface, creating lower pressure above the wing. This pressure difference produces lift without requiring any geometric tilt.

For practical applications, cambered airfoils are best suited for fixed-wing UAVs designed for long endurance and for gliders, as they provide high lift at low angles of attack and offer a favorable lift-to-drag ratio. This improves energy efficiency and allows longer flight durations. Symmetric airfoils are more suitable for racing drone propellers because they perform well at higher speeds, maintain aerodynamic balance, and can handle rapid changes in airflow direction effectively.

Conclusion:

Airfoils play a crucial role in aerodynamics by controlling how air flows around a surface to generate lift. The shape of an airfoil directly affects its performance in terms of lift and drag. Flat plate airfoils are the simplest form and can generate lift only when tilted, but they generally experience higher drag. Symmetric airfoils produce no lift at zero angle of attack but generate lift efficiently when tilted, making them suitable for control surfaces. Cambered airfoils, due to their curved shape, can generate lift even at zero angle of attack and produce higher lift at lower angles, which is why they are commonly used in aircraft wings. Overall, the choice of airfoil depends on the required lift, drag characteristics, and application.